London Underground Reliability API (1.0.0)

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A dual-model API tracking user-reported delays and official TfL live statuses with background auto-updating.

Root

Responses

> **200** Successful Response

GET /

Response samples

200

Content type
application/json

Copy

```
null
```

Create Report

REQUEST BODY SCHEMA: application/json
required

- `line_name` required
  string (TfLLine)
  Enum: `"Bakerloo"` `"Central"` `"Circle"` `"District"` `"Elizabeth line"` `"Hammersmith & City"` `"Jubilee"` `"Metropolitan"` `"Northern"` `"Piccadilly"` `"Victoria"` `"Waterloo & City"`
- `delay_minutes` required
  integer (Delay Minutes) (0 .. 300]
  Delay in minutes (1 to 300)
- `observed_experience` required
  string (ObservedExperience)
  Enum: `"Stuck in tunnel"` `"Train moving at a crawling pace"` `"Platform dangerously crowded"` `"Train cancelled/disappeared from board"` `"Not sure, just delayed"`

Responses

> **201** Successful Response

— **422** Validation Error (Pagination Limits Exceeded)

— **429** Rate Limit Exceeded

POST /reports

Request samples

Payload

Content type
application/json

Copy

```
{
    "delay_minutes": 25,
    "line_name": "Central",
    "observed_experience": "Train moving at a crawling pace"
}
```

Response samples

201 422 429

Content type
application/json

Copy

```
{
    "delay_minutes": 25,
    "edit_token": "550e8400-e29b-41d4-a716-446655440000",
    "id": 1042,
    "line_name": "Central",
    "observed_experience": "Train moving at a crawling pace",
    "report_date": "2026-03-09 17:25:00"
}
```

Get Reports

QUERY PARAMETERS

skip	integer (Skip) >= 0 Default: 0 Records to skip for pagination
limit	integer (Limit) [1 .. 100] Default: 50 Max 100 records per request to prevent DoS
line_name >	TfLLine (string) or Line Name (null) (Line Name)
experience >	ObservedExperience (string) or Experience (null) (Experience)

Responses

> 200 Successful Response

> 422 Validation Error

GET /reports

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
       "delay_minutes": 25,
       "id": 1042,
       "line_name": "Central",
       "observed_experience": "Train moving at a crawling pace",
       "report_date": "2026-03-09 17:25:00"
   }
]
```


Update Report

PATH PARAMETERS

report_id required	integer (Report Id)

HEADER PARAMETERS

x-edit-token required	string (X-Edit-Token) The secret token provided when the report was created

REQUEST BODY SCHEMA: application/json
required

line_name required	string (TfLLine) Enum: "Bakerloo" "Central" "Circle" "District" "Elizabeth line" "Hammersmith & City" "Jubilee" "Metropolitan" "Northern" "Piccadilly" "Victoria" "Waterloo & City"
delay_minutes required	integer (Delay Minutes) (0 .. 300] Delay in minutes (1 to 300)
observed_experience required	string (ObservedExperience) Enum: "Stuck in tunnel" "Train moving at a crawling pace" "Platform dangerously crowded" "Train cancelled/disappeared from board" "Not sure, just delayed"

Responses

> 200 Successful Response

— 403 Forbidden (Invalid Token)

— 404 Report Not Found

— 422 Validation Error (Invalid ID Type)

PUT /reports/{report_id}

Request samples

Payload

Content type
application/json

Copy

```
{
    "delay_minutes": 25,
    "line_name": "Central",
    "observed_experience": "Train moving at a crawling pace"
}
```

Response samples

200 403 404 422

Content type
application/json

Copy

```
{
    "delay_minutes": 25,
    "id": 1042,
    "line_name": "Central",
    "observed_experience": "Train moving at a crawling pace",
    "report_date": "2026-03-09 17:25:00"
}
```

Delete Report

PATH PARAMETERS

report_id required	integer (Report Id)

HEADER PARAMETERS

x-edit-token required	string (X-Edit-Token) The secret token provided when the report was created

Responses

— **204** Successful Response

> **422** Validation Error

DELETE /reports/{report_id}

Response samples

422

Content type
application/json

Copy Expand all Collapse all


```
{
  - "detail": [
      + { … }
    ]
}
```


Get Live Status

Responses

> **200** Successful Response

GET /live-status

Response samples

200

Content type
application/json

Copy Expand all Collapse all


```
[
  - {
        "id": 500,
        "line_name": "Central",
        "reason": "Signal failure at Stratford",
        "status": "Severe Delays",
        "timestamp": "2026-03-09 17:20:00"
    }
]
```


Get Live Status History

Fetches the recent history of official TfL statuses for a specific line.

PATH PARAMETERS

line_name required	string (TfLLine) Enum: "Bakerloo" "Central" "Circle" "District" "Elizabeth line" "Hammersmith & City" "Jubilee" "Metropolitan" "Northern" "Piccadilly" "Victoria" "Waterloo & City"

QUERY PARAMETERS

limit

integer (Limit) [1 .. 100]
Default: 20
Limit the number of historical records returned

Responses

> 200 Successful Response

> 422 Validation Error

GET /live-status/{line_name}/history

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
       "id": 500,
       "line_name": "Central",
       "reason": "Signal failure at Stratford",
       "status": "Severe Delays",
       "timestamp": "2026-03-09 17:20:00"
   }
]
```


Get Discrepancies

Responses

> 200 Successful Response

GET /analytics/discrepancies

Response samples

200

Content type
application/json

Copy Expand all Collapse all

```
[
 - {
        "confidence_level": "High (Confirmed by crowd)",
        "corroborating_reports": 8,
        "crowd_consensus_minutes": 22.5,
        "line_name": "Central",
        "official_status": "Good Service",
        "peak_delay_minutes": 45
    }
]
```


Get Network Uptime

Responses

> **200** Successful Response

GET /analytics/uptime

Response samples

200

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
        "current_status": "Good Service",
        "line_name": "Victoria",
        "official_disruption_snapshots": 3,
        "uptime_percentage": 98.5
    }
]
```


Get Health Check

Responses

> **200** Successful Response

— **503** System Degraded

GET /health

Response samples

200 503

Content type
application/json

Copy

```
{
    "api_uptime_seconds": 3402.5,
    "background_worker_status": "healthy",
    "database_status": "connected",
    "last_tfl_sync": "2026-03-09 17:30:00"
}
```

Get Delay Velocity

PATH PARAMETERS

`line_name` required

string (TfLLine)
Enum: `"Bakerloo"` `"Central"` `"Circle"` `"District"` `"Elizabeth line"` `"Hammersmith & City"` `"Jubilee"` `"Metropolitan"` `"Northern"` `"Piccadilly"` `"Victoria"` `"Waterloo & City"`

Responses

> **200** Successful Response

— **422** Validation Error (Invalid Line Name)

GET /analytics/velocity/{line_name}

Response samples

200 422

Content type
application/json

Copy

```
{
    "current_hour_delay_minutes": 45,
    "line_name": "Jubilee",
    "previous_hour_delay_minutes": 15,
    "trend": "+30.0 minutes",
    "velocity_assessment": "Accelerating (Peak severity currently at 45 mins)"
}
```

Get Reliability Score

PATH PARAMETERS

`line_name` required

string (TfLLine)
Enum: `"Bakerloo"` `"Central"` `"Circle"` `"District"` `"Elizabeth line"` `"Hammersmith & City"` `"Jubilee"` `"Metropolitan"` `"Northern"` `"Piccadilly"` `"Victoria"` `"Waterloo & City"`

Responses

> **200** Successful Response

— **422** Validation Error (Invalid Line Name)

GET /analytics/reliability/{line_name}

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "assessment": "Degraded (High Commuter Risk)",
    "line_name": "Central",
    "official_status": "Minor Delays",
    "reliability_percentage": 55,
  - "user_metrics": {
        "buffer_time_index": 22.5,
        "crowd_consensus_delay": 22.5,
        "peak_delay": 45,
        "total_reports": 14
    }
}
```


Get Delay Patterns

Responses

> **200** Successful Response

GET /analytics/patterns

Response samples

200

Content type
application/json

Copy Expand all Collapse all

```
[
 - {
        "average_delay_minutes": 28.5,
        "incident_count": 45,
        "observed_experience": "Stuck in tunnel",
        "peak_delay_minutes": 60
    }
]
```


Get Temporal Summary

Responses

> **200** Successful Response

GET /analytics/temporal

Response samples

200

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
        "day_of_week": "Monday",
        "hour_of_day": "17",
        "total_incidents": 120
    }
]
```